

Ahmet Halit Ünsal

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Executive Summary

Senior Engineering Leader & Chief Embedded Systems Architect with 15+ years professional experience and 8+ years directorship managing cross-functional teams up to 30 engineers. Deep technical mastery in Model-Based Design (MATLAB/Simulink/Stateflow), EV Powertrains (VCU/MCU/Inverters/BMS), PMSM FOC motor control, AUTOSAR, and ISO 26262 ASIL D compliance.

Tailored Competencies for TUSAŞ Motor Sanayii (TEI) (Head of Powertrain & ECU Embedded Software Architect)

- Core Technical Stack: Model-Based Design (MBD), MATLAB / Simulink / Stateflow, Physical Modeling & Dynamic Simulation, ISO 26262 (Functional Safety ASIL A-D), ASPICE Level 2/3 Process Implementation, UN R155 / UN R156 Automotive Cybersecurity (CSMS), AUTOSAR Classic (BSW/ASW/RTE), FreeRTOS & RTOS Middleware Architecture, C / C++ Bare-Metal & Low-Level Drivers, EV Powertrains: VCU, MCU, Inverters, OBC, BMS
- Product Showcase: EduTrace (<https://edutrace.net>) & AURA (<https://www.auratrading.org/>)

Professional Experience

Software Manager & NISO TECHNOLOGY CO

May 2021 - Present & Izmir, TR

Directed a multidisciplinary engineering organization of 30 software engineers across 4 team leads, delivering mission-critical embedded software for EV powertrain systems including ECUs, traction inverters, and on-board chargers (OBCs).

- Scaled organizational capability by establishing a team-lead layer, defining ownership boundaries, and running structured performance reviews and mentorship programs - reducing manager single points of failure and enabling parallel delivery across subsystems.
- Enforced ISO 26262 and SPICE compliance across BSW and ASW layers developed on FreeRTOS and other RTOS platforms, ensuring functional safety traceability from requirements through integration.
- Deployed CI/CD pipelines and automated tooling that compressed release cycle times and shifted regression detection left, reducing late-stage integration failures.
- Led UN R155/R156 cybersecurity standard implementation activities, aligning software development processes with vehicle cybersecurity management system (CSMS) requirements.
- Designed and deployed C-based middleware solutions for high-reliability embedded applications, and contributed directly to diagnostic and safety-critical algorithms for powertrain systems.
- Instituted Git-based version control strategies with branching policies that enforced traceability and streamlined cross-team collaboration across a large, concurrent codebase.

- Drove data-driven project management by deploying analytics tooling to surface velocity, defect density, and milestone risk metrics to stakeholders.
- Managed resource allocation and risk mitigation across competing program milestones, acting as the primary interface between software, hardware, and systems engineering teams.

Team Leader & ECEMTAG

November 2017 - May 2021 & Istanbul, TR

Led design and validation of model-based control algorithms for ECUs and TCUs across automotive and defense programs.

- Introduced automated testing frameworks and HIL simulation environments that raised software verification coverage and reduced physical prototype dependency.
- Developed safety and redundancy functions in compliance with military standards, coordinating requirements directly with defense customers.
- Acted as cross-functional integration lead, aligning software and hardware milestones across embedded, systems, and validation teams using JIRA-based program tracking.
- Contributed to E/E architecture process development and rollout, including tool qualification and development workflow standardization.

Lead Researcher & TUBITAK (Scientific and Technological Research Council of Turkey)

December 2016 - November 2017 & Kocaeli, TR

- Directed automotive and defense research programs focused on advanced control systems and diagnostics.
- Developed a product-level tool for data dictionary generation, calibration, and MiL (Model-in-Loop) testing, accelerating algorithm validation workflows.
- Characterized IPMSM performance in motoring and regenerative modes on a dynamometer test bench.

Senior Systems Engineer & TUMOSAN

January 2016 - December 2016 & Istanbul, TR

- Developed automatic shift-selection logic using Simulink Stateflow for a hybrid drivetrain platform.
- Designed traction control algorithms for slip-slide protection under variable surface conditions.
- Optimized fuel consumption and battery usage profiles using multi-objective cost functions.

Systems Engineer & METRO CO

February 2013 - January 2016 & Kocaeli, TR

- Built a longitudinal vehicle model and derived drivetrain power and torque requirements from first principles.
- Developed a full PMSM driver including Maximum Torque Per Ampere (MTPA) and flux-weakening control algorithms.

- Automated PID tuning via phase and gain margin calculation, and conducted HIL validation using dSpace MABX-I.

Researcher && Iowa State University - Bio-Electric Laboratory

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- Supervised advanced research on transcranial magnetic stimulation (TMS) and electric field modeling.
- Instructed electric machines coursework to sophomore engineering students.

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